

LITERATURE STATUS FOR ARXIV:1003.4261

AGENT_LANE_10

SOURCE QUESTION

In the Introduction of arXiv:1003.4261, Naor recalls the Cheeger–Kleiner–Naor lower bound that embeddings of the $n \times n \times n$ grid in the discrete 3-dimensional Heisenberg group into L_1 incur distortion at least $(\log n)^c$ for some universal $c > 0$. He then states that evaluating the supremum of such exponents remains an important open question, with $1/2$ being the largest possible value allowed by the known $\sqrt{\log n}$ upper bound.

ANSWER

Naor–Young, arXiv:2004.12522, *Foliated corona decompositions*, explicitly identifies the asymptotic evaluation of $c_{\ell_1}(\mathcal{B}_n)$ as a question asked in several earlier sources, including Naor’s ICM article [1]. Their Theorem 1.6 proves the sharp asymptotic

$$c_{\ell_1}(\mathcal{B}_n) \asymp \sqrt[4]{\log n} \quad (n \geq 2),$$

where $\mathcal{B}_n$ is the word ball in the discrete 3-dimensional Heisenberg group. Hence the exponent asked for in arXiv:1003.4261 is exactly $1/4$.

SCOPE

This is a literature-status answer for the Heisenberg distortion exponent question only. The later question in arXiv:1003.4261 about invariant metrics on cyclic groups and the dependence on the Abelian group exponent is separate and is not settled by this note.

REFERENCES

- [1] A. Naor, *L_1 embeddings of the Heisenberg group and fast estimation of graph isoperimetry*, arXiv:1003.4261.
- [2] A. Naor and R. Young, *Foliated corona decompositions*, arXiv:2004.12522.